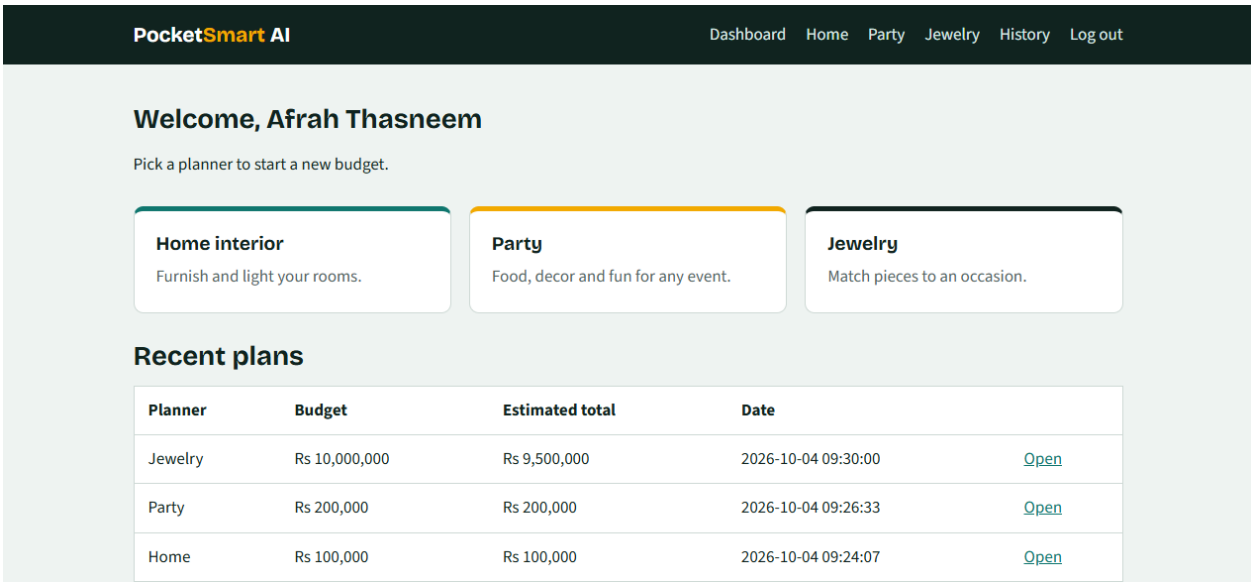
PocketSmart AI

Your Smart Budget & Recommendation Assistant

PROJECT DOCUMENT



Reference interface — PocketSmart AI dashboard.

A GenAI-powered budgeting and recommendation system for Home Interior Planning, Party Planning, and Jewelry Recommendations.

Technology focus: FastAPI, Gemini 1.5 Flash Pro, HTML/CSS/JavaScript, Jinja2, session-based personalization, and platform-oriented recommendations.

1. Introduction

Managing budgets across different lifestyle needs such as home decor, event planning, and jewelry shopping can be difficult because users must compare many products, services, platforms, and price ranges. PocketSmart AI is designed as a smart budget and recommendation assistant that turns a user’s budget and preferences into structured recommendations.

The project documentation describes a GenAI-powered recommendation workflow using Gemini 1.5 Flash Pro. The system accepts text, numerical budget information, and optional images for jewelry planning, and presents recommendations across platforms such as Amazon, Flipkart, IKEA, Swiggy, Zomato, and OYO.

The supplied reference screens demonstrate the working user experience: a dashboard, recent plans, category planners, budget allocation bars, recommendation cards, platform search links, and saving guidance.

1.1 Purpose

- Help users plan spending within a defined budget.
- Divide a budget into meaningful categories such as lighting, dining, decoration, catering, or jewelry.
- Provide recommendation cards with item names, estimated prices, and marketplace links.
- Support three main domains: Home Interior, Party Planning, and Jewelry.
- Maintain a user-friendly dashboard and history-oriented experience.

1.2 Project Scope

- Budget-based planning and recommendation generation.
- Authentication and session management.
- Planner-specific forms and result pages.
- Optional image input for jewelry recommendations.
- Fallback/default recommendations when AI suggestions are unavailable.
- Responsive presentation of recommendations using card-style UI.

2. Objectives

- Develop an intelligent budget assistant for everyday planning needs.
- Generate context-aware recommendations from user budgets and preferences.
- Use a multimodal AI model for text and optional image-based jewelry analysis.
- Create separate planning flows for Home, Party, and Jewelry categories.
- Provide platform-aware recommendations and external search links.
- Store or expose recommendation history and session-specific information.
- Provide fallback recommendations so that the application remains useful when AI suggestions are unavailable.
- Evaluate recommendation quality, budget adherence, and platform accuracy during testing.

2.1 Key User Scenarios

Home Interior: Users specify a budget, room details, and quantities for items such as lights, ceiling fans, and dining tables. The system allocates spending and recommends suitable products.

Party Planning: Users provide a budget, guest count, event type, and venue details. The system distributes spending across catering, decoration, entertainment, and venue-related needs.

Jewelry: Users specify budget, occasion, and style preferences. An outfit image can optionally be supplied so that recommendations can consider color coordination and aesthetics.

3. System Architecture

The project documentation describes a modular architecture in which the frontend communicates with a FastAPI backend. The backend connects planner routes and session/authentication logic with the Gemini-based recommendation layer. The frontend uses HTML/CSS/JavaScript and Jinja2 templates to collect inputs and display results.

3.1 Major Components

Component	Role
Frontend UI	Collects budgets, preferences, category details and optional images; displays recommendation
FastAPI Backend	Provides routing, request handling, authentication/session support and planner endpoints.
Gemini AI Layer	Interprets budget context, preferences and optional images and generates recommendation co
Recommendation Logic	Formats budget allocations, planner-specific prompts, recommendation structures and fallback
External Platforms	Marketplace/service destinations such as Amazon, Flipkart, IKEA, Swiggy, Zomato and OYO.
History & Session	Supports personalized experience and access to previous recommendation queries/results.

3.2 Backend Route Structure

- /generate-home — Home interior recommendation generation.
- /generate-party — Party planning recommendations.
- /generate-jewelry — Jewelry recommendations with text and optional image input.
- /register, /login, /logout, /token — authentication and session control.
- /session-info, /session-data — user/session metadata and personalization data.
- /recommendations-details — detailed recommendations for a selected category and budget.
- /history — access to past recommendation queries and results.
- /startup and application entry point — initialize services/configuration and start the server.

4. Technologies and Development Workflow

4.1 Core Technologies

- FastAPI: backend service, routing, API handling and session-related functionality.
- Gemini 1.5 Flash Pro: multimodal AI model described in the project documentation for budget interpretation, recommendation generation and image analysis.
- HTML/CSS/JavaScript: frontend presentation and interactive input/output handling.
- Jinja2: template-driven frontend rendering.
- Third-party platform links/APIs: sourcing or linking to products and services from multiple platforms.

4.2 Development Workflow

- Configure Google Cloud access and Gemini API connectivity.
- Validate both text-only and text-plus-image prompts.
- Build modular planner routes and recommendation services.
- Implement authentication, session handling and structured input/output.
- Develop planner forms and recommendation result pages.
- Add platform links and fallback/default recommendations.
- Test with varied budgets and realistic Home, Party and Jewelry scenarios.
- Optimize prompts, validation, session behavior and UI/UX.

4.3 Suggested Project Structure

The project documentation describes a modular structure containing route/end-point logic, service/recommendation logic, model/input-output schemas, templates, and the main backend application. The exact file names can vary with the implementation, but the documented separation is:

- routes/ — endpoint and planner routing logic.
- services/ — recommendation logic and AI calls.
- models/ — input/output schemas.
- templates/ — Jinja2 HTML pages.
- static/ — frontend CSS and JavaScript assets.
- main application — FastAPI initialization, CORS, startup and server entry point.

5. User Interface and Functional Screens

The following screens are the supplied reference photographs of the working PocketSmart AI interface. They are included as project documentation evidence.

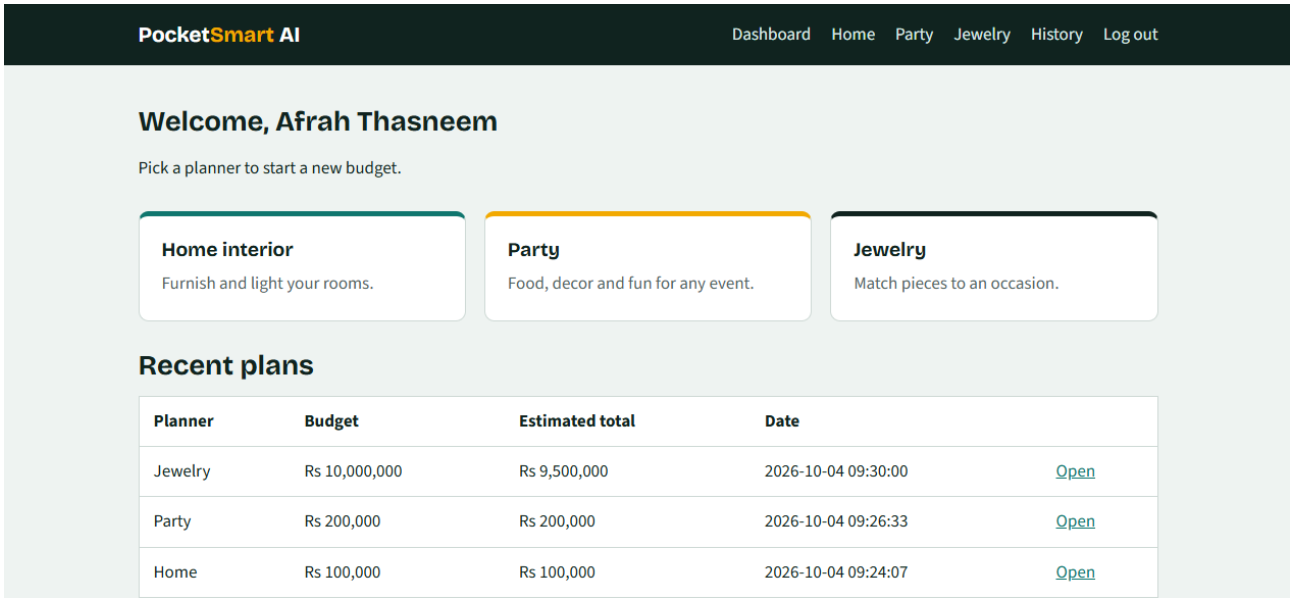


Figure 1. PocketSmart AI dashboard showing planner categories and recent plans.

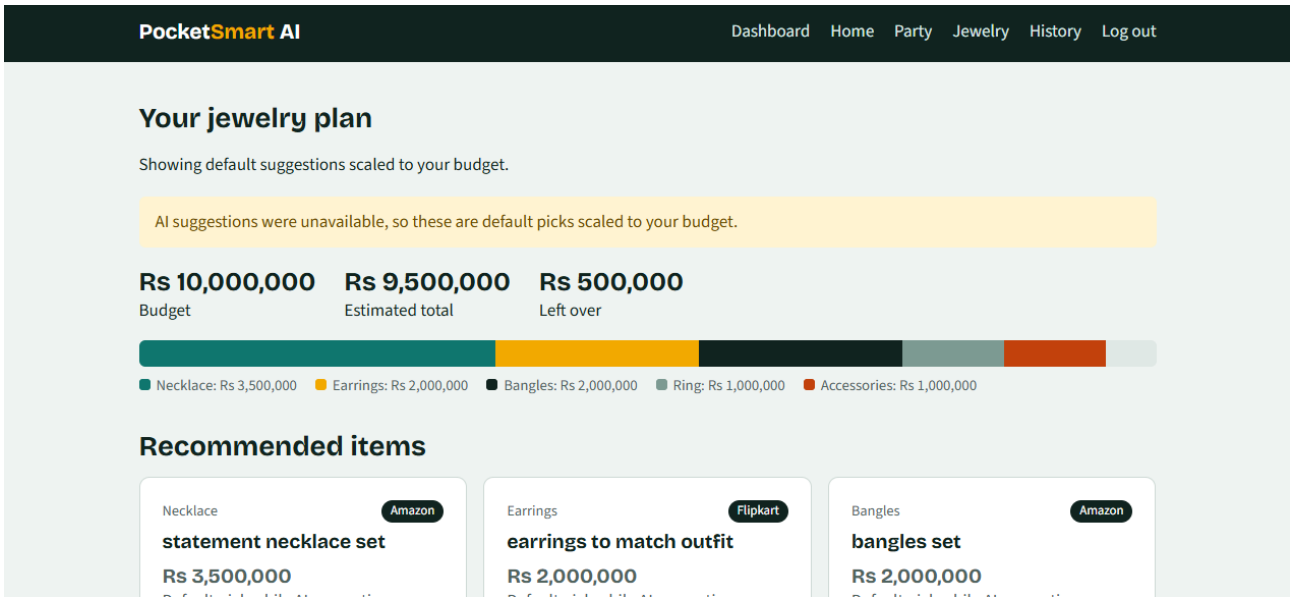


Figure 2. Jewelry recommendation plan with budget allocation and recommended items.

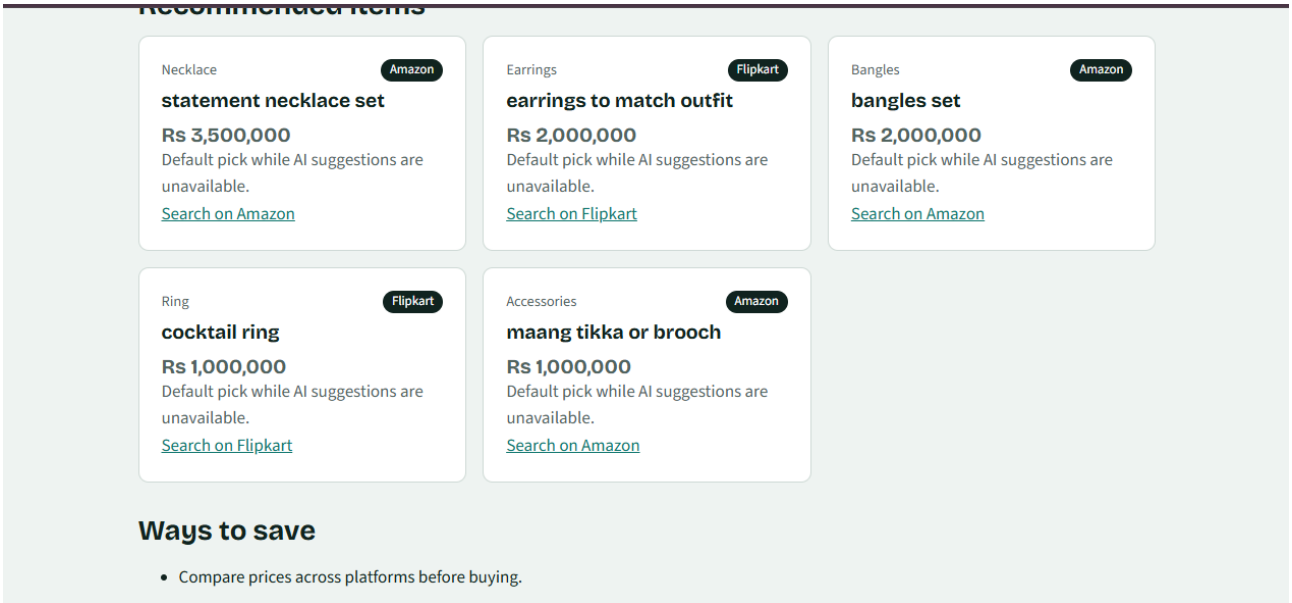


Figure 3. Jewelry recommendation cards and saving guidance.

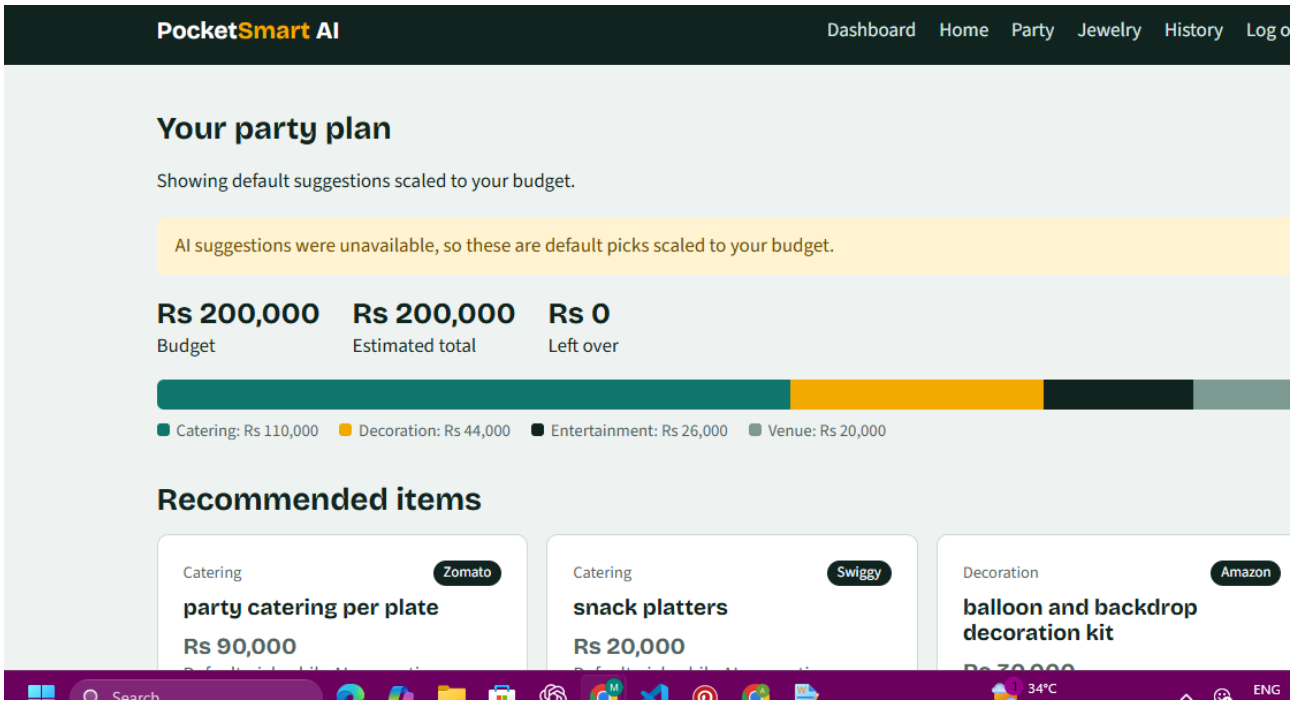


Figure 4. Party planner budget allocation and recommendation section.

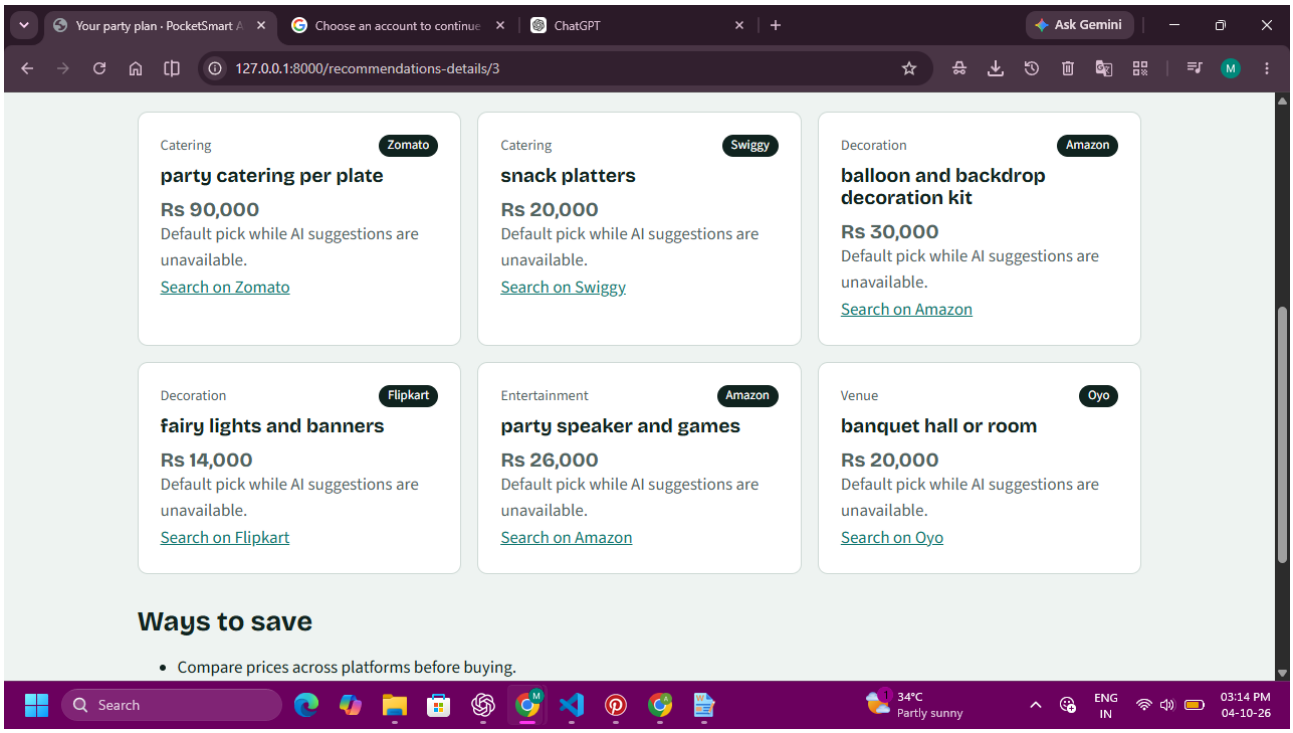


Figure 5. Party recommendation details with catering, decoration, entertainment and venue.

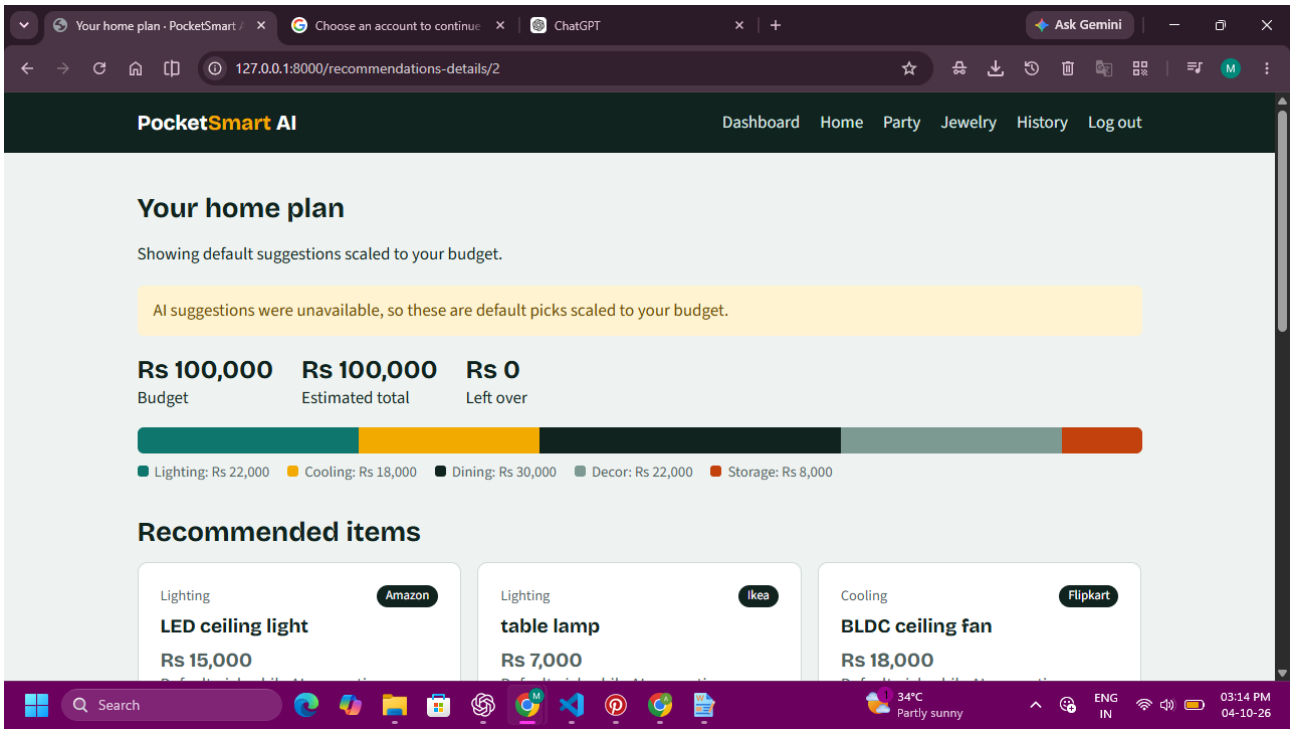


Figure 6. Home planner budget allocation and initial recommendations.

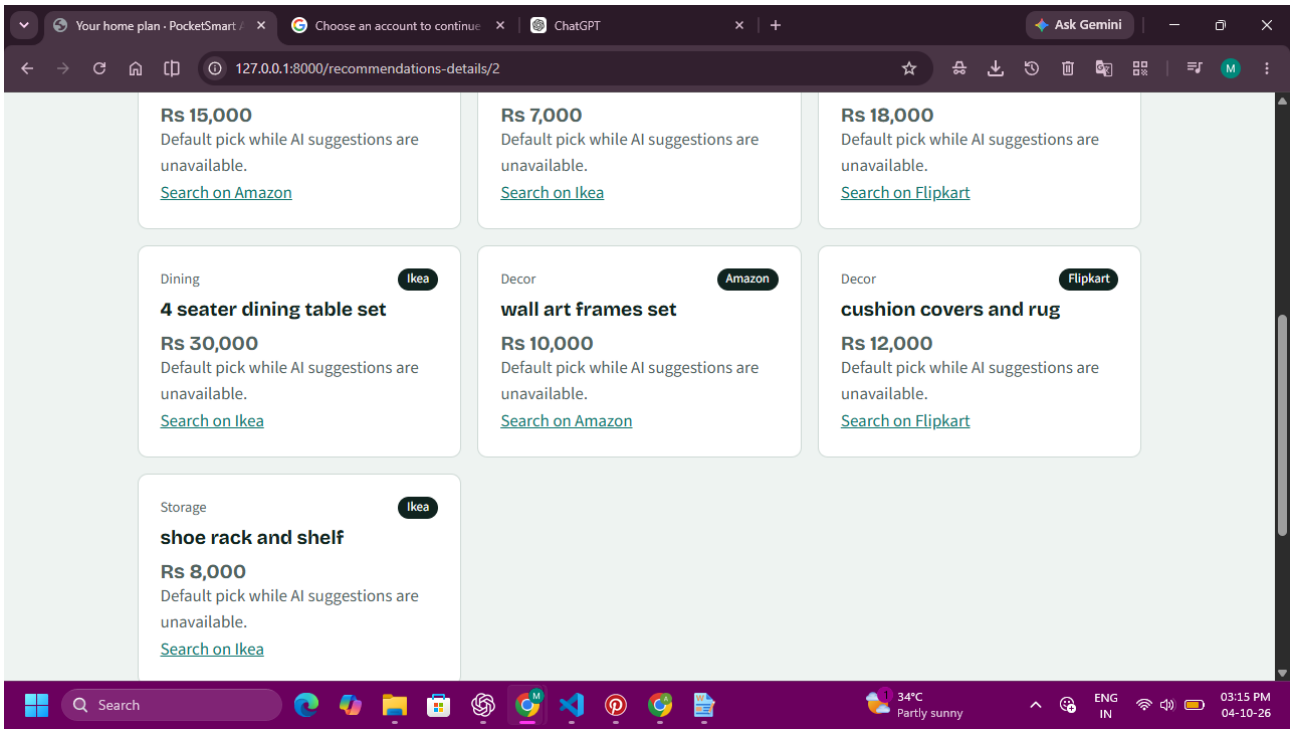


Figure 7. Home recommendation details with lighting, cooling, dining, decor and storage.

6. Functional Analysis of the Reference Screens

6.1 Dashboard

- The dashboard welcomes the signed-in user and presents three planner choices: Home Interior, Party, and Jewelry.
- A Recent Plans table displays planner name, budget, estimated total, date, and an Open action.
- The dashboard provides a central starting point for creating or revisiting budget plans.

6.2 Jewelry Module

- The jewelry page shows Budget, Estimated Total, and Left Over values.
- A horizontal budget allocation visualization divides spending among Necklace, Earrings, Bangles, Ring and Accessories.
- Recommendation cards include category, item title, estimated price, marketplace label, explanatory text and a search link.
- The reference screen explicitly displays a fallback notice stating that default suggestions are being shown because AI suggestions were unavailable.

6.3 Party Module

- The party page allocates the budget among Catering, Decoration, Entertainment and Venue.
- The displayed example shows a budget and estimated total of Rs 200,000 with Rs 0 left over.
- Recommendation cards cover catering, decoration, entertainment and venue, with marketplace/service links such as Zomato, Swiggy, Amazon, Flipkart and OYO.

6.4 Home Module

- The home page allocates the budget among Lighting, Cooling, Dining, Decor and Storage.
- The example shows a Rs 100,000 budget with a matching estimated total.
- Recommendation cards include items such as LED ceiling light, table lamp, BLDC ceiling fan, dining table set, wall art frames, cushion covers/rug, and shoe rack/shelf.

7. Budget Allocation and Recommendation Logic

PocketSmart AI is designed to make the relationship between a user’s budget and recommended items visible. The reference screens show a summary of the total budget, estimated total, and amount left over, followed by a category-wise allocation bar.

7.1 Example: Jewelry

- Budget shown in the reference: Rs 10,000,000.
- Estimated total: Rs 9,500,000.
- Left over: Rs 500,000.
- Displayed allocations: Necklace Rs 3,500,000; Earrings Rs 2,000,000; Bangles Rs 2,000,000; Ring Rs 1,000,000; Accessories Rs 1,000,000.

7.2 Example: Party

- Budget shown: Rs 200,000.
- Estimated total: Rs 200,000.
- Left over: Rs 0.
- Displayed allocations: Catering Rs 110,000; Decoration Rs 44,000; Entertainment Rs 26,000; Venue Rs 20,000.

7.3 Example: Home

- Budget shown: Rs 100,000.
- Estimated total: Rs 100,000.
- Left over: Rs 0.
- Displayed allocations: Lighting Rs 22,000; Cooling Rs 18,000; Dining Rs 30,000; Decor Rs 22,000; Storage Rs 8,000.

These values are examples visible in the supplied reference screenshots and are not presented here as universal recommended spending ratios.

8. Authentication, Sessions and History

The project documentation includes authentication and personalization features. Registration creates a user account, login authenticates an existing user, and logout terminates the active session. A token endpoint is described for authorized access to protected API routes.

- Register: creates a new user account.
- Login: authenticates existing credentials and starts a session.
- Logout: ends the current session and redirects to login.
- Token: provides a secure JWT token for authorized API access.
- Session Info: returns current user/session metadata.
- Session Data: supports personalized session-specific recommendation information.
- History: displays past recommendation queries and results for review or reuse.

9. Testing and Optimization

The documented testing milestone focuses on reliability, recommendation quality, contextual awareness, budget adherence, platform accuracy, validation, session handling and responsive UI/UX.

- Test multiple budgets across Home, Party and Jewelry.
- Test varied user preferences and planning contexts.
- Check whether estimated totals stay within the provided budget.
- Validate platform links and category matching.
- Test text input and optional image input for jewelry.
- Check fallback/default recommendations when AI output is insufficient.
- Validate edge cases and improve prompts for more consistent outputs.
- Check the interface across different screen sizes.

10. Results and Observations

The supplied screenshots demonstrate that the application can present a complete budget-planning flow from dashboard selection to detailed recommendations. The interface communicates budget totals, category allocations, recommendation cards, external marketplace/service search links, and saving guidance.

An important observed behavior is the fallback mechanism. When AI suggestions are unavailable, the interface clearly informs the user and displays default picks scaled to the selected budget. This supports continuity of the user experience instead of leaving the recommendation page empty.

The three modules also demonstrate domain-specific recommendation structures: home planning emphasizes household categories; party planning emphasizes catering, decoration, entertainment and venue; jewelry planning emphasizes accessory categories and occasion/style-oriented selection.

11. Advantages

- Budget-centered planning instead of unstructured shopping.
- Multiple lifestyle domains in one application.
- AI-based personalization described in the project documentation.
- Optional multimodal input for jewelry planning.
- Clear category-wise budget visualization.
- Recommendation cards are easy to scan.
- External platform search links make recommendations actionable.
- Fallback recommendations improve resilience when AI suggestions are unavailable.

12. Limitations and Future Enhancement Areas

- The reference screens show fallback/default recommendations when AI suggestions are unavailable; stronger AI availability and error handling can improve this experience.
- External marketplace/service data may require reliable API integration or updated sourcing mechanisms.
- Recommendation quality depends on prompt design, available product/service information and user input.
- Future versions can improve price freshness, product availability checks, filtering, ranking and personalization.
- The system can be extended with richer analytics, saved budgets, notifications and more planning categories.

13. Conclusion

PocketSmart AI presents a practical application of generative AI for budget-aware everyday decision-making. The project combines a web interface, FastAPI backend, Gemini-based recommendation workflow, authentication/session features, and category-specific planning modules.

The reference implementation demonstrates Home Interior, Party, and Jewelry planning experiences with budget summaries, category allocations, recommendation cards and marketplace/service links. By combining personalization with visible budget control and fallback recommendations, the system aims to save users time and simplify planning.

The project documentation concludes that the system brings intelligent recommendations to users through a budget-friendly, fast and personalized interface.

14. References / Project Source Basis

This project document was prepared from the supplied PocketSmart AI project documentation and the seven reference screenshots provided with the request. The documentation describes the project purpose, architecture, technologies, routes, milestones, UI modules, testing approach and conclusion.

Primary source: PocketSmart AI: Your Smart Budget & Recommendation Assistant — supplied project document.

Reference visuals: seven screenshots supplied in the conversation, covering Dashboard, Jewelry, Party and Home planner/recommendation screens.